COMPSCI 311: Introduction to Algorithms **Fall 2026**

Discussion 6

Your Name: _____ Collaborators: _____

You will be randomly assigned groups to work on these problems in discussion section.

Problem 1. Recurrence. Consider the recurrence $T(n) = T(n-1) + \sqrt{n}$ with $T(1) = 1$. Use unrolling to show that $T(n)$ is $\Theta(n^d)$ for some $d > 0$.

Problem 2. Divide and Conquer the Tree Consider the following algorithm to compute a MST for a connected weighted graph $G = (V, E)$: partition V into two subsets differing in size by at most 1; recursively find the MSTs for these two "halves"; join these two MSTs by the smallest-weight edge across the cut. Does this give a valid algorithm? Prove or give a counterexample. What is the complexity?

Problem 3. Proof by Induction for Recurrences. Consider the following recurrence, which describes an algorithm that divides a problem of size n into two equal-sized subproblems, but then does $O(n \log n)$ outside of the recursive calls:

$$T(n) \leq 2T(n/2) + cn \log n,$$

We again assume the recurrence holds for $n \geq 2$ and that $T(2) \leq c$. Prove by induction that $T(n) \leq cn(\log n)^2$. (Another way to say this is to say that $T(n)$ is $O(n \log^2 n)$.) You should assume that the logarithm is base 2, so that $\log(n/2) = \log n - 1$.

Hint: see p. 213 of the book and slides from earlier this week. The algebra for this proof is slightly longer but follows the same pattern.